

GOONJAN SAHA

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Education

Indian Institute of Technology Delhi
B.Tech. Computer Science & Engineering

2020 – 2024
CGPA: 9.03/10

Honors and Awards

Conference Reviewer – Peer-reviewed 8 papers at MaGMAR@ACL'26, SCALE@ICML'26, SD4H@ICML'26 *Apr'26-May'26*
AI Specialist, Samsung Electronics – Internal certification for **advanced AI** engineering and research **proficiency** *Dec'24*
IITD Endowment Merit Scholarship – Recognized among the **top 30** students for exceptional academic performance *Sep'22*
Department Change to CSE – Secured **3rd position** among all applicable first-year students to merit a department transfer *Jul'21*
Olympiads and National Exams – KVPY-SX **AIR 191**, JEE Advanced **AIR 1213**, INMO qualifier, NSEA top-300 *2018-2020*

Publications

MediTOD: High-Quality Medical History-Taking Dialogue Dataset (EMNLP 2024) 🎧 *New Delhi, India*
Co-Author at EMNLP'24 | Supervisor: Prof. Mausam 🎧 *May'23 – May'24*

- Co-built the **first public** English medical Task-Oriented Dialogue dataset (**22,503** utterances, novel **CMAS attribute schema**)
- Led the **data-curation pipeline**: collaborated with **clinicians** on a questionnaire-based **labeling** scheme; designed an **attribute schema** capturing complex symptom-attribute links; **canonicalized** to the UMLS Metathesaurus for downstream interoperability
- Fine-tuned Llama3, BioGPT, Flan-T5** baselines for NLU, Policy, NLG subtasks; establishing medical-dialogue **benchmarks**

Work Experience

Samsung Electronics HQ *Suwon, South Korea*
AI Engineer | Visual Display Business Unit *Sep'24 - Present*

- Agentic Movie Metadata Chatbot — LangGraph, ReAct, Postgres+pgvector** *Apr'26 – Present*
 - Architected a **ReAct-style agentic chatbot** for natural-language queries over **5,000+** multi-version movie metadata corpus
 - Redesigned** streaming PostgreSQL schema into **hybrid** relational + pgvector + denormalized views to upgrade latency, accuracy
 - Designed and validated 5-point **LLM-as-a-Judge** on held-out **persona-stratified synthetic-generated** data on each agent step
 - Achieved **4.6** Answer Relevance, **4.9** Completeness, **4.7** Logical Planning on held-out evals; TTFT under **6s** for the full agent loop
- Multi-Agentic Pipeline for Non-biometric Character Recognition** *Oct'25 – Jan'26*
 - Designed **multi-agentic pipeline** that resolves character under non-biometric **constrained** setting of **no audio, no facial** data
 - Delivered **85%** average/**92%** peak precision in character attribution, unblocking movie-tagging pipeline obeying legal constraints
- Heuristic-based Subtitle-Frame Temporal Alignment** *Feb'25 – Mar'25*
 - Engineered a **data-cleaning and alignment** pipeline to sync 1 FPS video frames with noisy subtitles streams **distorted by ads**
 - Achieved **100% alignment accuracy** (bounded by 1s granularity) across **30** production assets using a 3-stage cascading pipeline

Selected Projects

KeyArrange: MP3 → Playable Piano Music Sheet | *Hobby Project (ongoing) 🎧*

- Audio-to-playable-piano-arrangement pipeline; **stem separation, transcription, arrangement** engine; deployed on HF Spaces

Select Projects Completed at IIT Delhi | *Course Projects*

- Linux Kernel** modules · Multi-cycle ARM-ISA **processor** · WHILE-program **compiler** · Adversarial-search **Game AI** agent

Technical Skills

ML / AI	PyTorch, HuggingFace, LangGraph, LoRA (PEFT), Seq2Seq fine-tuning, RAG, ReAct Agents
Others	C/C++, Python, Java, Bash, Git, AWS (EC2, S3, DynamoDB), Docker, PostgreSQL, pgvector

Leadership / Extracurricular

Engineering Mentor | *Incoming Intern, Samsung Electronics HQ* *May'26 – Present*

- Onboarding lead and **task scoping** for an incoming engineering intern; **evaluating** performance as a prospective full-time hire

Find Your Co-Founder | *Entrepreneurship Development Cell, IIT Delhi* *Jan'23*

- Participated in a **Networking** session; **pitched myself** as a Resilient Technical Co-founder; **brainstormed** ideas with teammates